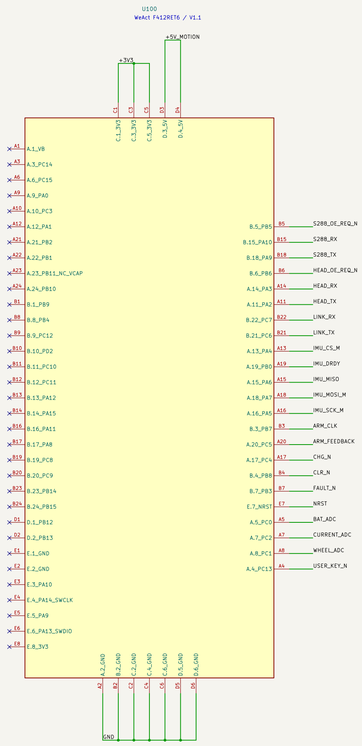
MORI / MOTION CARRIER / S2

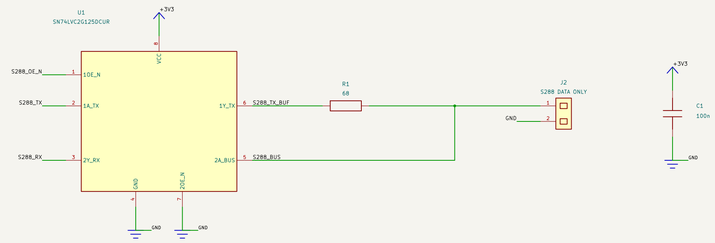
Manually composed circuits. Same electrical interfaces. PROTOTYPE / UNVALIDATED.

01 STM32 module / GPIO map



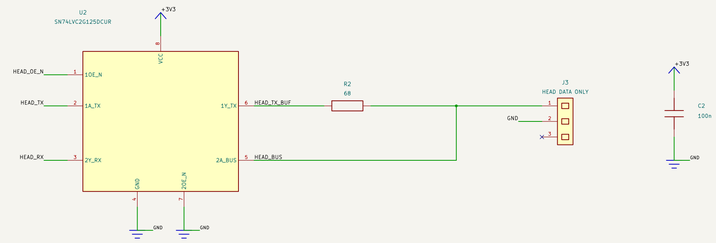
NC pins are intentionally unused. SWD probe connects on the module top.

02 Wheel TTL / 6 Mbps target



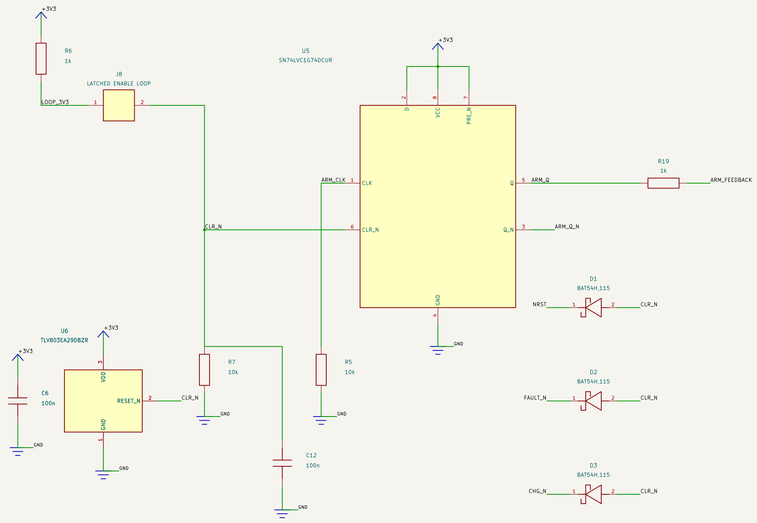
3.3 V logic; actuator thresholds and timing remain bench items.

03 Head TTL / separate bus



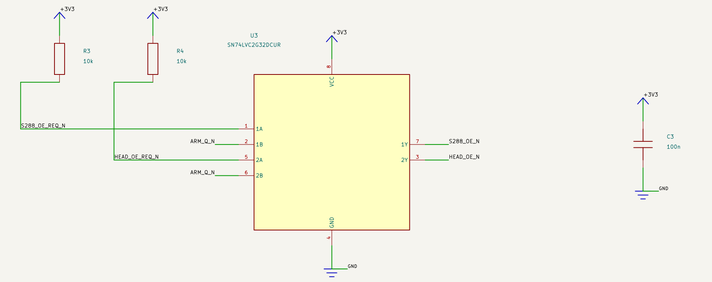
3.3 V logic; actuator thresholds and timing remain bench items.

04 Explicit ARM latch / physical disable loop



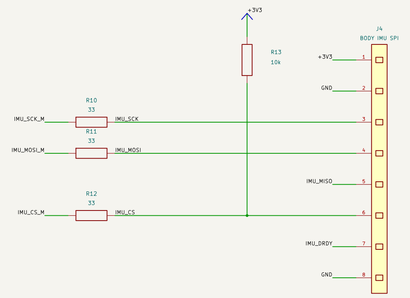
CLR_N dominates. Re-arm requires a new ARM_CLK edge: this is not a watchdog.

05 Transmit-enable gating



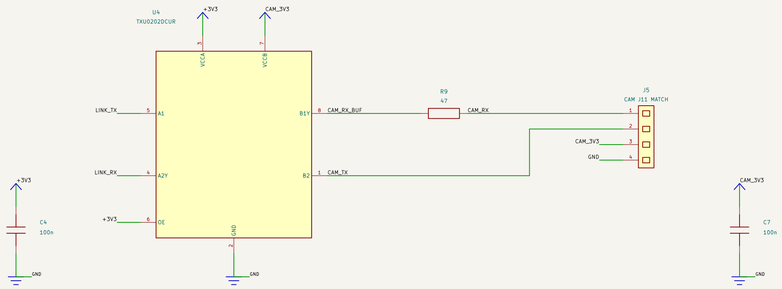
OE_N = request_N OR ARM_Q_N. Pull-ups disable TX by default.

06 Body IMU connector / source damping



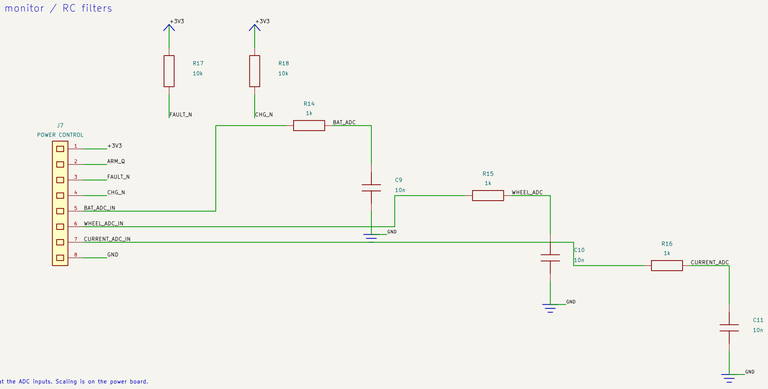
SPI and DRDY belong to the rigid body IMU, not the head.

07 Interaction UART / power-domain isolation



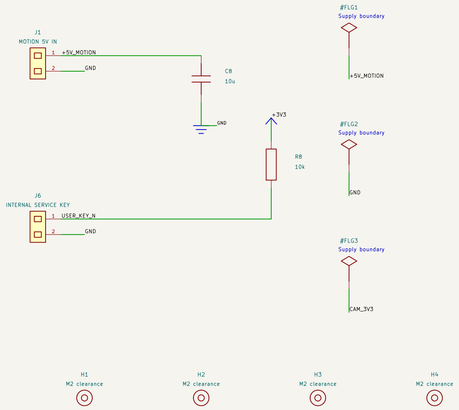
CAM_3V3 powers translator B only. It does not power the motion MCU.

08 Power monitor / RC filters



1k / 10nF filters at the ADC inputs. Scaling is on the power board.

09 Input / user button / supply boundaries



J6 is a user button. Physical disable is J8: reset can cause a fall.

PROTOTYPE / NOT_TESTED — circuit and pin assignments retained

Sheet: /
File: MORI_motion_P5R2.kicad_sch

Title: MORI / motion / P5 — FUNCTIONAL SCHEMATIC

Size: User | Date: 2026-09-23 | Rev: V1.2-H0.5-P5

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